

MIT-ADT AI Grand Challenge 2026

- **Problem Statement ID –Healthcare-3**
- **Problem Statement Title-AI System to Detect Malnutrition**
- **Theme- Healthcare**
- **Category- Software**
- **Team Name – Git Push Pray**

Health Byte AI — Smart Food Label Intelligence

- Proposed Solution

Proposed Solution: NutriScan AI is an AI-powered web app that transforms any packaged food label scan into actionable nutritional intelligence. A user photographs or uploads a product label; the system:

Extracts ingredients via OCR

Maps them to nutritional values

Detects deficiencies against personalised RDA thresholds

Tracks dietary patterns over 7 days

Generates AI-powered explanations and trend predictions (via Claude API)

Recommends healthier alternatives

Problem Anchor: India's NFHS-5 data shows 57% of women aged 15–49 and 67% of children under 5 are anaemic primarily due to iron deficiency. This is an *information and choice problem*, not a food scarcity problem.



TECHNICAL APPROACH



Technologies Used:

7-Layer Pipeline:

- 1. Input — User uploads label image or uses manual entry fallback (OCR with confidence score)
- 2. Extraction — LLM parses OCR output into structured JSON
- 3. Analysis — Nutrition Estimator (Open Food Facts) + User Profile (age, gender, goal)
- 4. Detection — Deficiency Detector (compares against personalised RDA)
- 5. Tracking — 7-day Dietary Pattern Engine (trend direction: improving / worsening / stable)
- 6. AI Insight Layer (*NEW v3.0*) — Claude API/Open AI generates natural language explanations, trend summaries, predictive risk warnings
- 7. Output — Recommendation Engine (rule-based swaps + AI insights → final card)

Layer	Technology
Frontend	React (web app)
OCR	Tesseract.js / Google Vision API
LLM – Parsing	Anthropic Claude API (claude-sonnet-4)/Open AI
LLM – Insights	Anthropic Claude API (claude-sonnet-4)/Open AI
Nutrition DB	Open Food Facts API + local CSV fallback
Recommendation	Python rule-based engine + Nutri-Score + AI insights
Backend	FastAPI (Python)
Hosting	Vercel (frontend) + Render (backend)



FEASIBILITY AND VIABILITY



Feasibility:

- Built on top of an existing ingredient scanner (Sarvesh's existing project) — Phase 1 achievable in 2–3 weeks
- AI Insight Layer is a single Claude API call per session — lightweight and controllable
- OCR fallback (manual product name entry) prevents demo failure from poor label quality
- Modular architecture: each of the 7 layers is independently testable

Challenge	Risk	Mitigation
OCR accuracy on Indian labels	High	Two-path input; confidence score < 70% triggers user verification
Open Food Facts coverage of Indian brands	High	Priority 0: curated CSV of 50 hand-verified Indian products built first
Demo stability	Medium	Pre-loaded demo account with 5 scans; AI insights pre-visible
LLM output consistency	Low	Pre-loaded demo account with 5 scans; AI insights pre-visible



IMPACT AND BENEFITS

Target Audience: Urban Indian consumers (18–45), parents, fitness-conscious individuals, healthcare workers doing community-level dietary screening.

Social Impact:

- Addresses the core finding of NFHS-5: 57% of Indian women and 67% of children under 5 are anaemic
- Zero nutritional literacy required
AI explains deficiencies in plain language
- Predictive risk warnings before deficiency becomes serious

Evaluation Axis	NutriScan v3.0 Advantage
Innovation	Only system combining OCR + Nutri-Score + AI pattern detection + natural language insights + longitudinal tracking + product recommendation in one scan flow
AI Usage	LLM used twice: ingredient parsing + insight generation; returns structured JSON — explainable, not a black box
Impact	57% of Indian women anaemic (NFHS-5); explains deficiencies and predicts risk in plain language
Feasibility	Modular; Phase 1 achievable in 2–3 weeks
Scalability	Open Food Facts: 3M+ global products; add a country by updating one RDA config file

1) NFHS-5 Report

Official NFHS-5 Phase-II key findings PDF from Ministry of Health and Family Welfare, confirming anaemia stats (57% women, 67% children).

https://mohfw.gov.in/sites/default/files/NFHS-5_Phase-II_0.pdf

2) Nutri-Score Details

Peer-reviewed evaluation of the Nutri-Score algorithm (negative/positive points formula).

<https://pmc.ncbi.nlm.nih.gov/articles/PMC9421047/>

Wikipedia overview with computation rules:

<https://en.wikipedia.org/wiki/Nutri-Score>

3) Open Food Facts API

Official API documentation for product nutrition data (ingredients, nutrients per 100g).

<https://openfoodfacts.github.io/openfoodfacts-server/api/>

4) ICMR RDA 2020

ICMR-NIN RDA report (protein, iron, calcium thresholds for Indians).

Brief note PDF: https://www.nin.res.in/rdabook/brief_note.pdf

Full version (via Scribd): <https://www.scribd.com/document/526073382/RDA-2020-Full-Ver>

5) Research Paper

S. Janokar et al., "Food Ingredient Analysis Integrating OCR and LLM for Enhanced Consumer Safety," ICSCSA 2025, pp. 594-599.

doi: 10.1109/ICSCSA66339.2025.11171116

Keywords: OCR, LLM, Food Safety, AI in Healthcare, Ingredient Analysis